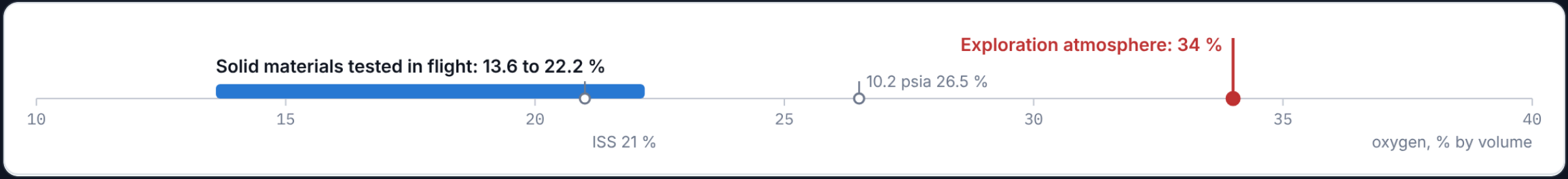
FlameGap

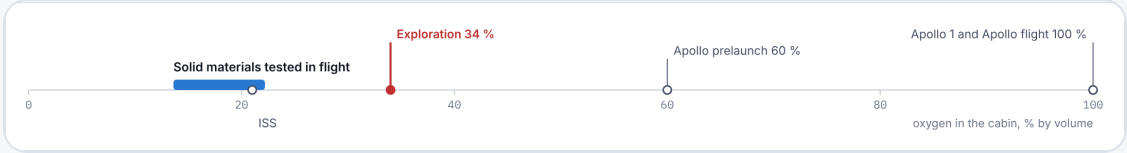
Where the flight evidence on spacecraft fire ends, and what it means for a Moon base



NASA plans to fly lunar crews in 34 % oxygen at 8.2 psia. Has anything been burned in that air?

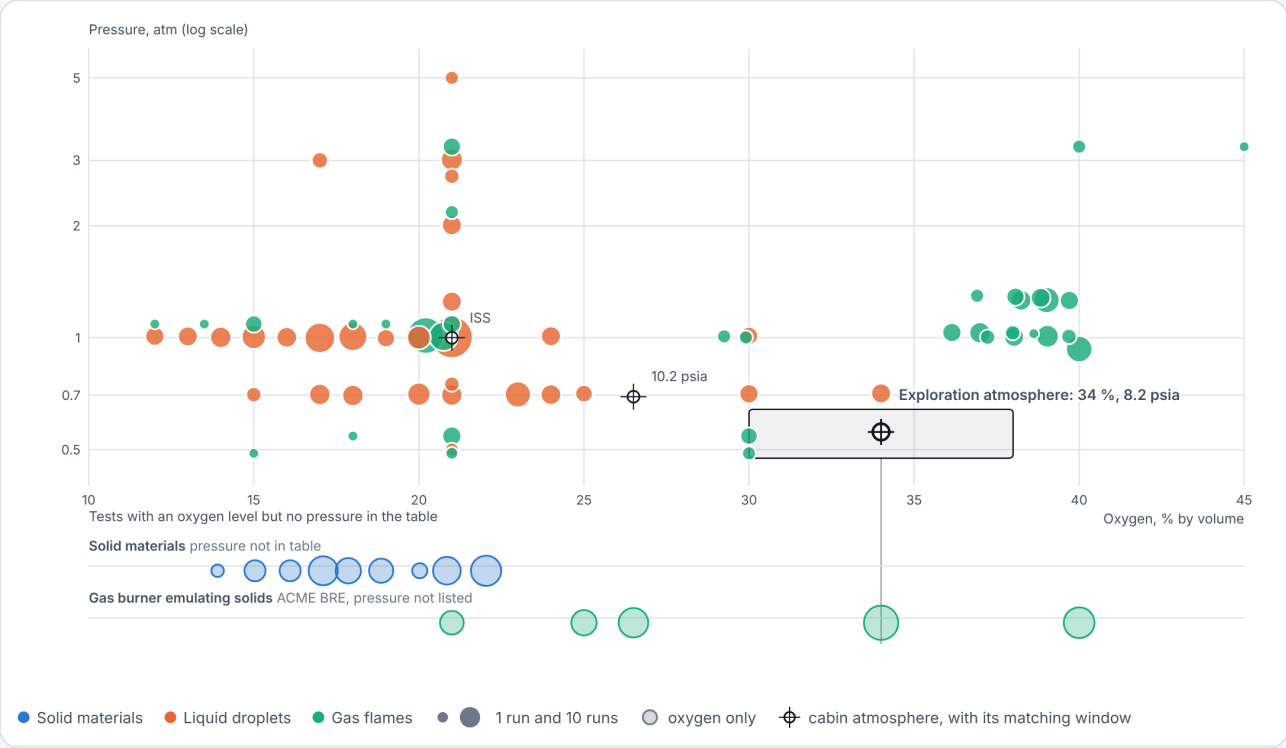
We read all **24 combustion investigations** in NASA's PSI archive: **986 test runs**, 352 papers and 6.3 TB of raw data, and placed every run by the oxygen and pressure it burned in.

> “Because of the pure oxygen atmosphere, the fire spread quickly…”
>
> Apollo 1, NASA Office of Safety and Mission Assurance



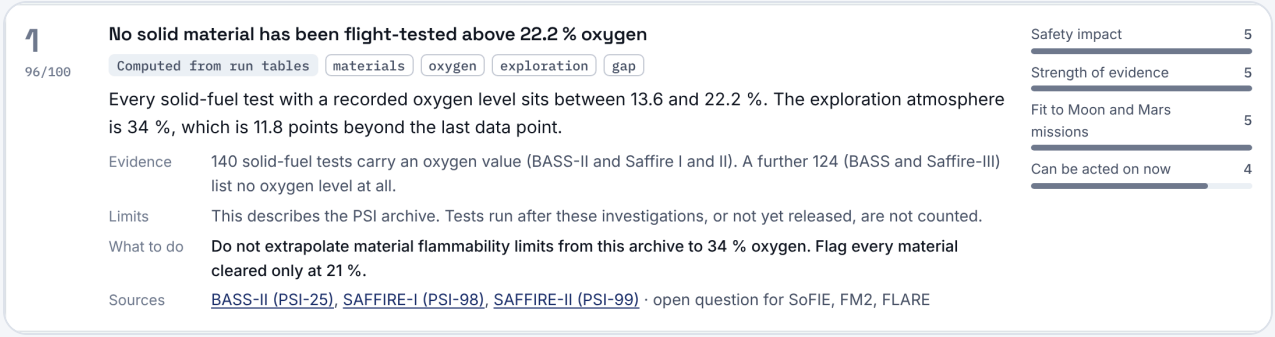
Cabin oxygen through history against the range of solid-material flight tests.

No solid material was flight-tested above 22.2 % oxygen



- Zero solid or droplet runs inside the exploration window.
- Droplets reached 34 %, but at **0.70 atm**, not 0.56.
- At 25 % oxygen and above, only **8 %** of droplet flames went out on their own, against 92 % below 18 %.
- Every run was in microgravity. None at lunar gravity.

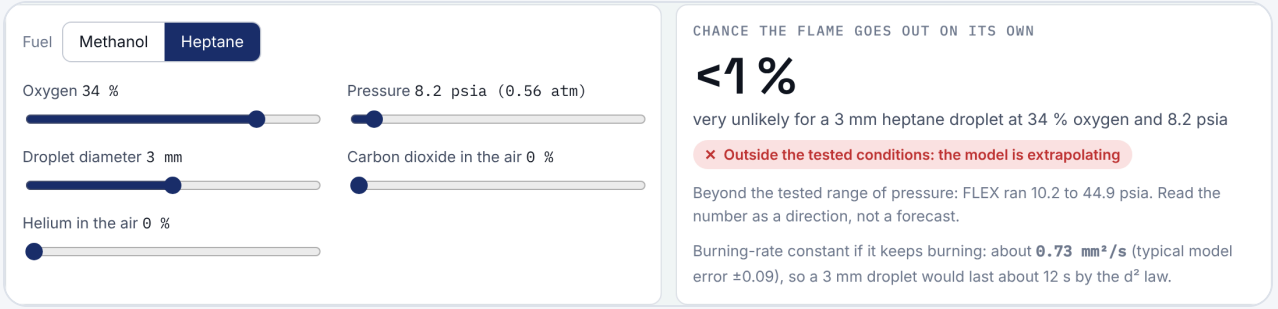
15 findings, each with evidence, limits, an action and NASA sources



Users weight **safety, evidence, mission fit and actionability**; the ranking follows. Top of the list:

- Cutting the airflow may not put a fire out in oxygen-rich air.
- NASA's own results say the material approval test is not the worst case.
- A local extinguisher jet can feed a fire in weightlessness.

Will a burning droplet go out on its own? And can we trust the answer?

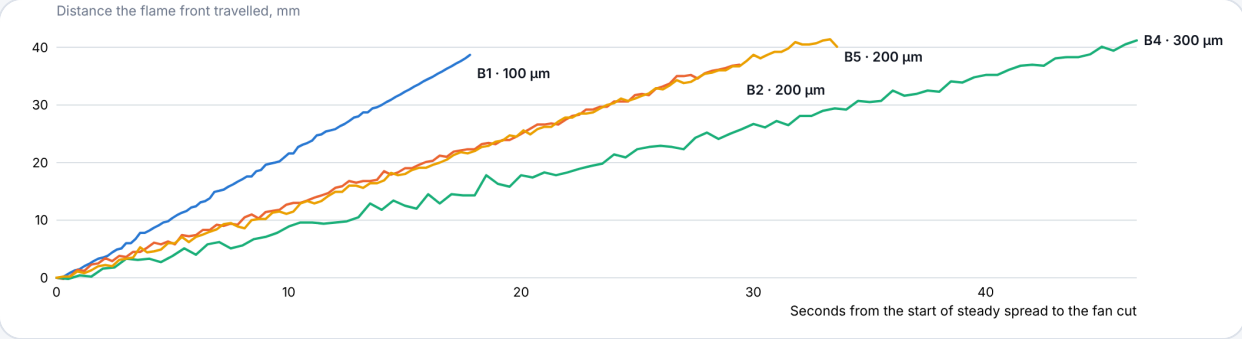


82 %
right on unseen test days

69 %
always guessing

Logistic model on **261 FLEX runs**, cross-validated by test date. It shows the nearest real runs and **flags extrapolation**: the exploration atmosphere is outside every FLEX pressure.

NASA keeps the results as raw video. We turned it into numbers.



5 / 5
independent checks pass

1.6 %
between repeat tests

- Flame out **1.9 s** after the fan cut; NASA report: within 2 s.
- Spread falls with film thickness, as thin-fuel theory predicts.
- Ready to run over all 244 BASS-II videos.

BUILT TO BE TRUSTED

Every number computed from NASA data. Every quote checked. Every AI claim verified.

- The build recomputes all figures from the raw PSI tables and **fails** if a quote is not word for word in its NASA source.
- **29 tests** recount key facts from the CSVs with independent code.
- Claude answers questions from the archive only; code checks each claim's records and numbers.

Next

- Feed the gaps to SoFIE and FM2, which can test elevated oxygen and lunar gravity.
- Publish a BASS results table from the videos.
- On Earth: the same science made firefighter air tanks lighter and fire detectors smarter.